

Module 04 Lab - Exploratory Data Analysis **Objective:** To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model. **This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading ****What is Exploratory Data Analysis (EDA)?**** EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns # Seaborn is a library built on top of Matplotlib that makes creating beautiful plots easier.

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("--- First 5 Rows ---")
print(df.head())

print("--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many non-null values are in each column, and their data type.
# Notice that 'Age' and 'Cabin' have missing values!
df.info()
```

```
--- First 5 Rows ---
  PassengerId  Survived  Pclass \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

   Name                               Sex  Age  SibSp \
0  Braund, Mr. Owen Harris             male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1
2    Heikkinen, Miss. Laina             female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)     female  35.0    1
4   Allen, Mr. William Henry             male  35.0    0

   Parch  Ticket   Fare Cabin Embarked
0      0    A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0    113803  53.1000  C123      S
4      0    373450   8.0500   NaN      S

--- Basic Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   PassengerId     891 non-null    int64
1   Survived        891 non-null    int64
2   Pclass          891 non-null    int64
3   Name            891 non-null    object
4   Sex             891 non-null    object
5   Age             714 non-null    float64
6   SibSp           891 non-null    int64
```

```

7   Parch      891 non-null   int64
8   Ticket     891 non-null   object
9   Fare       891 non-null   float64
10  Cabin      204 non-null   object
11  Embarked    889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

Part 2: Descriptive Statistics Let's start by getting a high-level numerical summary of the data.

- ✓ The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe())
print("--- Key Insight")
```

Part 3: Visual EDA - Telling Stories with Plots Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights. **A Note on**

- ✓ **Plotting Libraries:**
 - * **Matplotlib:** The foundational library, gives you full control over everything.*
 - Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

- ✓ Visualization 1: How many survived? A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

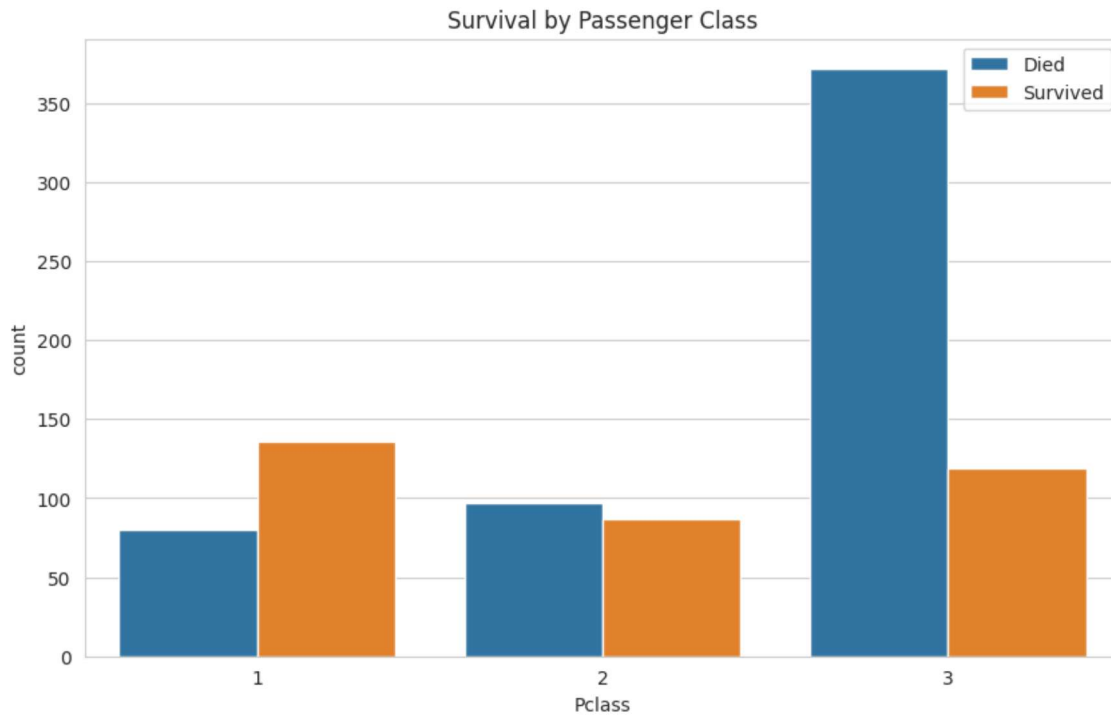
```
sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=d
```

- ✓ Visualization 2: Does passenger class matter for survival? Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

```

plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of s

```

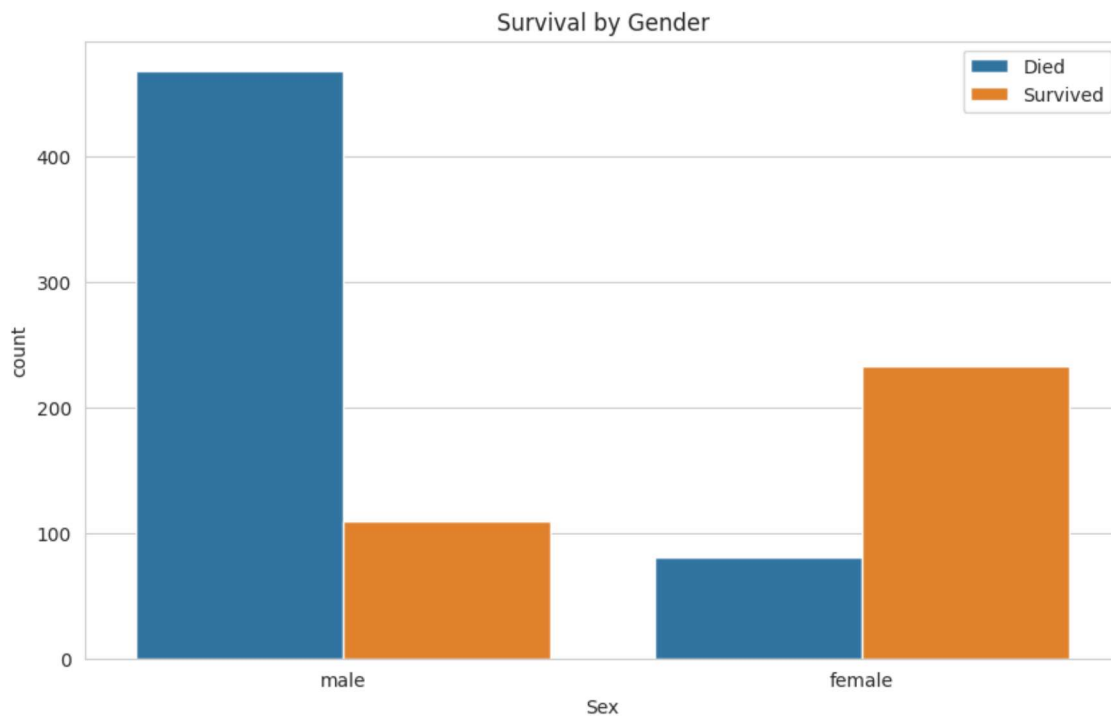


Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3rd class passengers.

Visualization 3: What about gender? The 'women and children first' mantra is famous. Let's see if the data supports it.

◆ Gemini

```
plt.figure(figsize=(10, 6))sns.countplot(x='Sex', hue='Survived', data=df)plt.title('Survival by Gender')
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: The pattern is undeniable. A much higher proportion of females survived compared
```



Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is another very strong

- Visualization 4: How does age play a role? For a continuous variable like `Age`, a histogram is a great way to see its distribution.

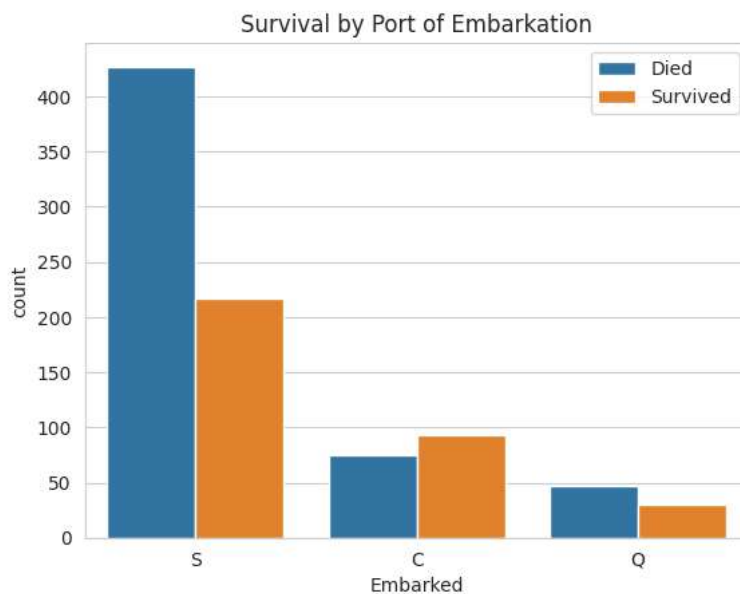
A FacetGrid allows us to create multiple plots side-by-side to compare distributions. # Here, we create one histogram for pass

- Part 4: Student Experimentation **Instructions:** Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation. The `Embarked` column tells you where a passenger boarded the ship (C =

- Cherbourg, Q = Queenstown, S = Southampton). 2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

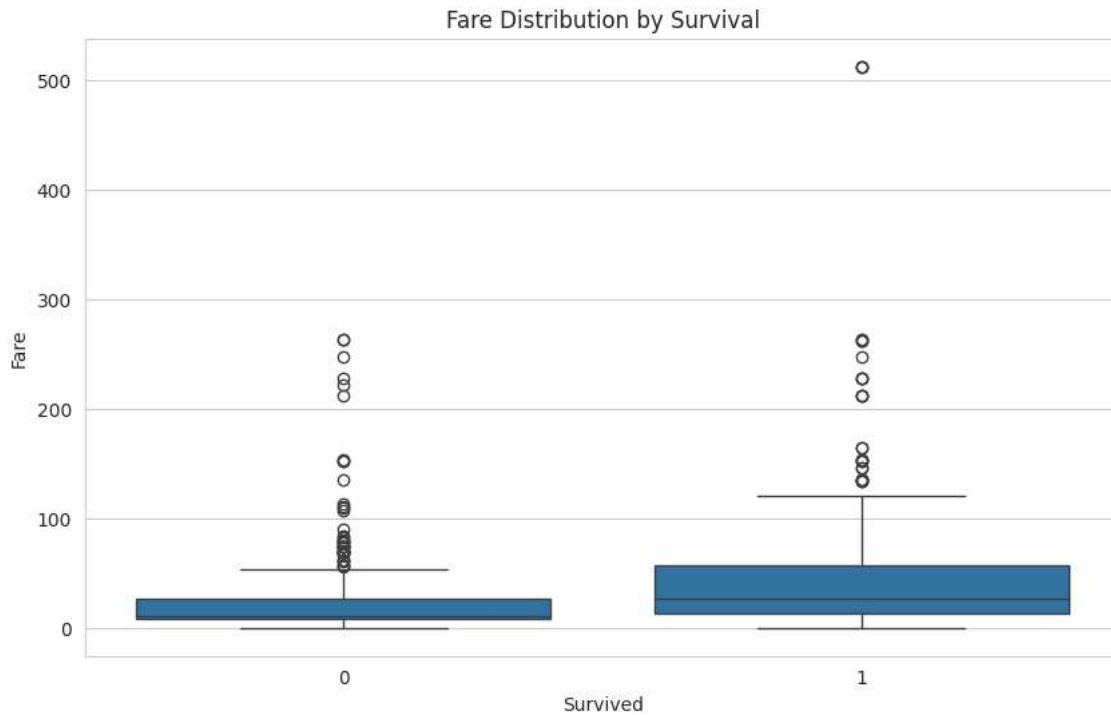
val rate compared to those who boarded at Southampton (S) or Queenstown (Q). This may be linked to class – Cherbourg had a higher



Insight: Passengers who boarded at Cherbourg (C) had a noticeably higher survival rate compared to those who boarded at Southamp

- Experiment 2: Fare vs. Survival. 1. `Fare` is a continuous numerical variable. 2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't. 3. Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you?

is wider for survivors. This lines up with the class pattern – passengers who paid more (often 1st class) were more likely to si



Insight: Passengers who survived tended to have paid higher fares on average, and the spread of fares is wider for survivors. Th

Close

Knowledge Check**Instructions:** Answer the following questions in this markdown cell.1. **What is the primary goal of Exploratory Data Analysis (EDA)?**2. **Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?** (Describe them in terms of class, gender, and age).3. **Why is it important to visualize data instead of just looking at summary statistics?** What can a plot show you that a number like 'mean' or 'count' can't?1 The primary goal of EDA is to understand the structure, patterns, and quality of a dataset before building a model – checking for missing values, spotting relationships between variables, and identifying trends or anomalies that will inform how the data should be cleaned and modeled.2. Based on the plots in this lab, the person with the best chance of survival was a 1st class young child or a woman in her 20s-30s, since class and gender were the strongest predictors (1st class and female passengers survived at higher rates) and young children also showed a notable survival spike compared to other age groups.3. It is important to visualize data instead of just looking at summary statistics because numbers like mean or count can hide important patterns, since two datasets can have the same mean but completely different distributions (skewed, bimodal, with outliers, etc.), and visualizing these shapes, relationships between variables, and outliers at a glance is a way that's much harder to catch from numbers alone.Elquin Ponce

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Knowledge Check**Instructions:** 1 The primary goal of Exploratory Data Analysis (EDA) is to understand the structure, patterns, and quality of a dataset before building a model,

checking for missing values, spotting relationships between variables, and identifying trends or anomalies that will inform how the data should be cleaned and modeled. 2. Based on the plots in this lab, the person with the best chance of survival was a 1st class, female, young child or a woman in her 20s-30s, since class and gender were the strongest predictors (1st class and female passengers survived at much higher rates) and young children also showed a notable survival spike compared to other age groups. 3. It is important to visualize